

Atividade 1 - Utilize as técnicas de leitura **Skimming** (leitura rápida para entender a ideia geral) e **Scanning** (leitura focada para localizar informações específicas) para realizar as atividades abaixo.

Depois de responder às perguntas, leia o texto novamente com mais atenção. Identifique palavras que você não conhece e crie um **vocabulário**, contendo:

- Palavra em inglês
- Significado em português

The Modern age of computers

It was in the period after World War II that major developments in computer technology and industry took place. Dr. John von Neumann developed the stored program concept near the end of the war. This concept revolutionized the computer industry because now the computer was controlled by a program kept inside the machine instead of being controlled manually with the changing of wires. Maurice V. Wilkes, a student at the University of Pennsylvania during the time von Neumann, Eckert and Mauchly developed their work, took the idea of the stored program computer, the EDAC. Two years later, in 1951, Mauchly and Eckert completed another stored program computer, the EDVAC. They formed their own company, the Remington-Rand Corporation, and developed the first commercial computer, the UNIVAC, which was delivered to the U.S. Census Bureau. However, by 1956, IBM, had replaced their company as the leading computer manufacturer.

These early computers are known as the first generation of computers. They used vacuum tubes to control their operations. These tubes were large, and the computer needed lots of them. Consequently, these computers occupied a large area and consumed a lot of power. They generated a great deal of heat and needed very good air conditioning. Furthermore, they were

too slow, and their memory capacity was small. On top of these problems, they broke down frequently and required constant maintenance and repair.

The second generation of computers began in 1947 at Bell Laboratories. J.Bardeen, W.H.Brattain, and W.Shockley solved the problems of vacuum tubes by developing the transistor, which won them the Nobel Prize. These computers were smaller, faster, and more reliable than the first generation's. Besides, they cost less power, and generated less heat. These computers were programmed with high-level programming languages like FORTRAN.

Around 1965, the third generation of computers appeared on the market. Thus, transistors were replaced by integrated circuits. These tiny integrated were smaller and more dependable than the second generation's transistors. Microchips (tiny boards that contain these circuits) were less than 1/8 of an inch square. Third generation computers were smaller, faster, more powerful, and more reliable than the previous generations'. In addition to their advantages, they were also less expensive.

Microminiaturization greatly reduced the size of the integrated circuits of fourth generation computers. Thousands of tiny circuits now fit onto a single chip, a rectangular or square piece of silicon, usually from 1/10 to 1/4 inch. Several layers of an integrated circuit are etched or imprinted upon this silicon piece, the chip. These computers can complete approximately 1,000,000 instructions per second and are 50 times faster than third generation computers.

Based on these facts, we can predict that today's computers will soon be obsolete. Japanese and American scientists are working on the "fifth generation project". Not only will these "supercomputers" be faster, cheaper, more powerful and more reliable than the one we use today, but it is also expected that they will be able to reason and communicate with humans in natural language. Although the prediction that these computers would be ready before the year 2000 was not fulfilled, advances in computer technology

proceed so quickly that very soon we will be considering our present state-of-the-art technology totally outdated. (Adaptado de Torres, 2001)

- A. Quando aconteceram os principais avanços tecnológicos dos computadores?
- B. Qual foi o conceito desenvolvido por John von Neumann?
- C. Qual a empresa líder na fabricação de computadores em 1956?
- D. Quando começou a segunda geração de computadores?
- E. Qual o objetivo geral do texto?

2 - Utilizando as estratégias de leitura, relacione os títulos aos textos:

- a) Sugarcane: The Economic Cycle
- b) The original “Engenhos”
- c) Decline in the 19th Century
- d) A clean, renewable fuel
- e) Flex-fuel: pure ethanol or mixed with gasoline
- f) Sugarcane today

() In 1975, the Brazilian government launched its National Alcohol Program, known as Proalcool, which diversified the output of the sugar industry. Significant investments were made, with support from the World Bank, to allow for the expansion of areas cultivated with sugarcane and the introduction of ethanol distilleries. Amid the worldwide petroleum crisis, the experience helped reduce Brazil’s vulnerability and increase energy security.

() By the 1800s, Brazil, once the world’s biggest sugar producer, fell to fifth spot with only 8% of world production. With the end of the “Coffee Cycle” in the 20th Century, there was a resumption of sugarcane cultivation to produce

sugar for the internal market. The states of São Paulo and Rio de Janeiro became main suppliers for Southern Brazil, bringing about a decline for the sector in the Northeast.

() “Engenhos”, or sugar mills, were run much like sugar factories. Their infrastructure included the “Big House”, which served as residence for the Lord of the operation, his family and servants; the Chapel, for Christian religious celebrations; the “Senzala”, where slaves were kept; and the “Engenho” itself, composed of different structures dedicated to various phases of sugar production.

() As of late 2007, sugarcane fields occupied about 7.8 million hectares in Brazil, or about 2% of all arable lands available in the country. This makes Brazil the number one producer of sugarcane in the world, followed by India, Thailand and Australia. Main production regions are South-Central Brazil, where close to 90% of overall production is concentrated, and the Northeast. There are two harvests per year, which allows Brazil to produce sugar and ethanol year round for both the internal market and for export.

() Five centuries ago, sugar was worth almost as much as gold throughout Europe, because its production was limited to quantities that couldn't meet the demand. That made growing sugarcane a very profitable enterprise, which couldn't be pursued in Europe primarily because of inadequate climate conditions.

() Falling demand for hydrated ethanol (E100) was compensated by an increase in the use of anhydrous ethanol mixed with gasoline, thanks to the expansion of Brazil's light vehicle fleet. By then, in 25 years of large-scale use of ethanol, Brazil had developed engine technologies and distribution logistics that were unprecedented in the world.